

DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.

+3V3_MCU FB1 120R @100MHz +3V3_USBHS_A

USB-004: USB-HS analog supply / FB1 + C43
FB1: 120 ohm at 100 MHz; DCR 0.05 ohm initial max.
Screen with combined USB-HS current: I = 55.3 mA.
Vdrop = IR = 2.765 mV; P = I²R = 0.153 mW.
After-test DCR 0.10 ohm: 5.53 mV / 0.306 mW.
C43: 10 uF +/-10%, 35 V X7R; initial 9.11 uF.
TDK nominal bias curve: 9.64 uF at 3.3 V.
Not transient or all-corners qualification: verify rail ramp.
Full calculations: ../design/usb_interface.md [USB-004]
ERC: #FLG03 declares the +3V3_MCU feed through FB1.
This does not qualify the pending upstream SYS_AON source.

SDRAM_A[0..12] SDRAM_BA[0..1] SDRAM_DQM[0..3] SDRAM_DQ[0..31]

GPIO P0 / P1 NOR_DQ_HOST0 NOR_DQ_HOST1 NOR_DQ_HOST2 NOR_CS_N NOR_INT_N DS0_PWR_REQ

GPIO P2 / P3 SDRAM_DQ2 SDRAM_DQ1 SDRAM_DQ0 QPOWER_BUTTON_N QSENS_INT1 QVOL_UP_N QPAGE_PREV_N QPAGE_NEXT_N QVOL_DOWN_N

GPIO P4 / P5 SDRAM_A4 SDRAM_A5 SDRAM_A6 SDRAM_A7 SDRAM_A8 SDRAM_A9 SDRAM_A10 SDRAM_A11 CAM_SDA CAM_SCL

GPIO P6 / P7 RADIO_SCLCLK RADIO_CIPD RADIO_COPIQ RADIO_CS_NQ SDRAM_DQ31 SDRAM_A12 SDRAM_DQ7 SDRAM_DQ12 SDRAM_DQ13 SDRAM_DQ14 SDRAM_DQ15 SDRAM_DQM0 SDRAM_DQM2

GPIO P8 / P9 NOR_DQ_HOST5 NOR_DQ_HOST6 NOR_DQ_HOST7 NOR_CS_NQ NOR_CHK_HOSTQ SDRAM_CLK_Q

GPIO PA / PB SDRAM_A3 SDRAM_A2 SDRAM_A1 SDRAM_A0 SDRAM_DQM3 SDRAM_DQM1 SDRAM_CKE SDRAM_WE_NQ SDRAM_CAS_NQ SDRAM_RAS_NQ SDRAM_DQ8 SDRAM_DQ9 SDRAM_DQ10 SDRAM_DQ11

GPIO PC / PD SDRAM_DQ30 SDRAM_DQ29 SDRAM_DQ28 SDRAM_DQ27 SDRAM_DQ26 SDRAM_DQ25 SDRAM_DQ24 SDRAM_DQ23 SDRAM_DQ22 SDRAM_DQ21 SDRAM_DQ20 SDRAM_DQ19 SDRAM_DQ18 SDRAM_DQ17 SDRAM_DQ16 SDRAM_BA1

CMS-016: SCIO_SCK=P601; CIPO=P602; COPI=P603; CS_N=P604.
READY=P706/OQ26; HANDSHAKE=P705/OQ19.
P706=RESET_REQ_N; P707=PWR_EN. Host-domain signals.
Timing, switched-domain isolation and reset arbitration remain open.

USB / MIPI AVCC_USBHS P814/USB_DP P815/USB_DM USBHS_DM USBHS_DP USBHS_RREF VCC_USB VSS_USB VSS1_USBHS VSS2_USBHS

AVCC_MIPI MIPL_CLN MIPL_CL_P MIPL_DLO_N MIPL_DLO_P MIPL_DLL_N MIPL_DLL_P VCC1B_MIPI VSS_MIPI

CM5-013: R2 VCC1B_MIPI = +1V8_MIPI from U23.
C111: 100 nF local bypass to VSS_MIPI/GND
(RABP1 QDG Rev.1.10, Table 1); C39 bypasses AVCC_MIPI.
Camera lanes and operating sequence remain pending.
Keep CSI/DSI stopped until integration is qualified.

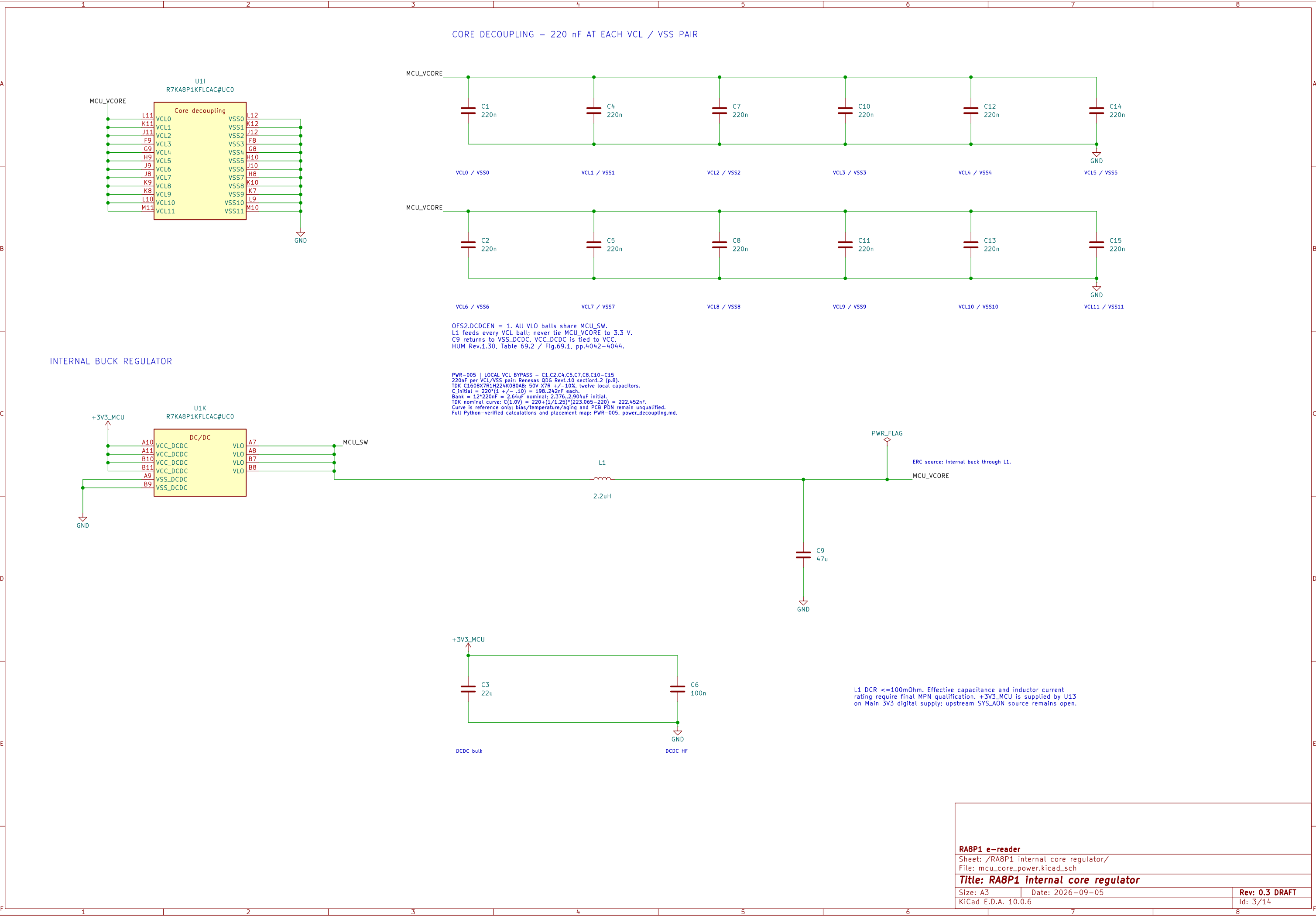
USB-001: R6 USBHS current-reference resistor
Renesas QDG Table 2, p.7: 2.2 kohm +/-1%.
Rmin = 2200*(1+0.01) = 2178 ohm
Initial tolerance only; not a derived PHY current.
Full basis + Python check: design/usb_interface.md

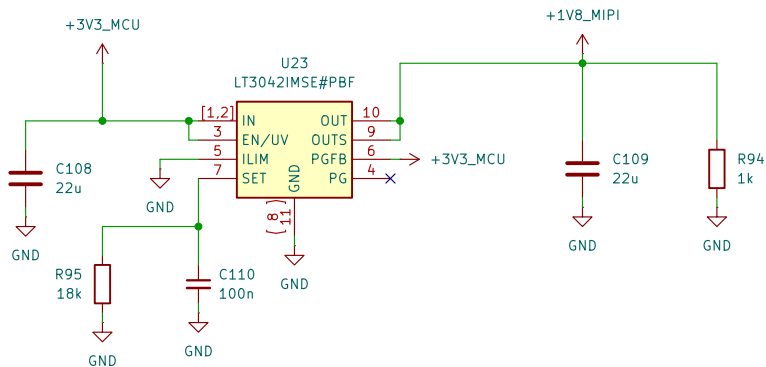
USB-002: USB full-speed supply / U1,U17, C40.
Renesas QDG Table 1, p.6: 3.3 V with 100 nF bypass.
C40 = 100*(1 +/-0.10) = 90.110 nF initial: 50 V X7R.
Nominal DC-bias estimate: 99.45 nF at 3.3 V (PWR-001).
Place C40 close to VCC_USB with a short VSS_USB return.
Full basis + Python check: design/usb_interface.md

USB-003: USB-HS supply / U1,H15, C41, C42
Renesas QDG Table 1 p.6: 10 nF ceramic + 47 uF electrolytic.
C41 = 10*(1 +/-0.10) = 9.11 nF initial: 50 V X7R.
TDK nominal bias model: C41(3.3 V) = 10.084 nF (not guaranteed).
C42 = 47*(1 +/-0.20) = 37.656 uF initial: 6.3 V.
Voltage screening: 3.6/6.3*100 = 57.14%; rating, not rail sign-off.
C42 positive to 3V3, negative to GND; no reverse voltage.
C41 close to H15; short returns to VSS1/VSS2_USBHS.
Full math + Python / qualification: ../design/usb_interface.md

PWR-001: C39 AVCC_MIPI bypass, 100 nF (QDG Table 2, p.7).
TDK C1608X7RH104K080AA; also used at C6 and C16-C34.
Initial C = 100*(1 +/-0.10) = 90.110 nF; 50 V X7R.
TDK DC-bias samples: 99.5575 nF / 3.15 V; 98.9525 nF / 4 V.
C(3.3 V) = 99.5575 + (3.3-3.15)/0.85*(-0.605) = 99.45 nF.
Interpolated nominal reference, NOT a guaranteed minimum.
Final rail / PDN qualification open; 3.3/50*100 = 6.6% rating.
Full basis + Python check: design/power_decoupling.md

RABP1 e-reader
Sheet: ./RABP1 IO allocation/
File: mcu_interfaces.kicad_sch
Title: RABP1 IO allocation
Size: A2 Date: 2026-09-05 Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.6 Id: 2/14





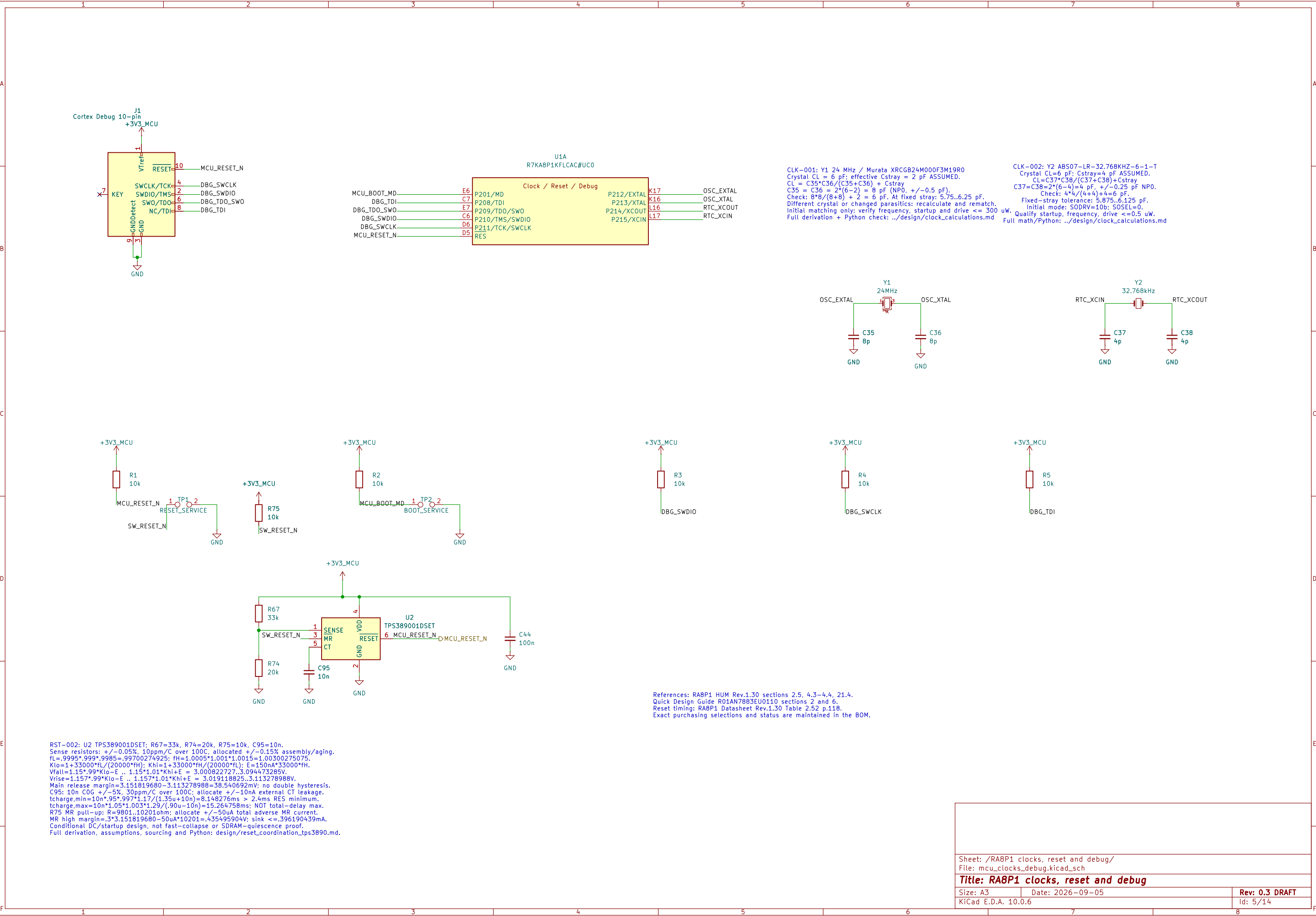
The diagram illustrates a PCB layout with four parallel power planes, each featuring a series of decoupling capacitors connected to ground. The planes are labeled as follows:

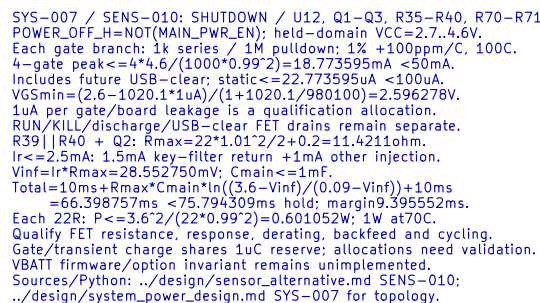
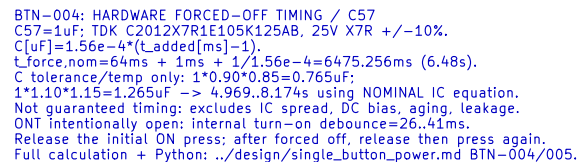
- Plane 1 (Top):** Labeled $+3V3_MCU$ at the input. It contains capacitors C16, C20, C24, C28, and C31, all with a value of 100n. The ground connection is labeled GND.
- Plane 2:** Labeled VCC_01 through VCC_05 . It contains capacitors C17, C21, C25, C29, and C32, all with a value of 100n. The ground connection is labeled GND.
- Plane 3:** Labeled VCC_06 through VCC_10 . It contains capacitors C18, C22, C26, C30, and C33, all with a value of 100n. The ground connection is labeled GND.
- Plane 4 (Bottom):** Labeled $+3V3_MCU$ at the input. It contains capacitors C19, C23, C27, and C34, all with a value of 100n. The ground connection is labeled GND.

The layout also includes labels for the power planes: VCC_01 , VCC_02 , VCC_03 , VCC_04 , VCC_05 , VCC_06 , VCC_07 , VCC_08 , VCC_09 , VCC_10 , $VCC2_11$, $VCC2_12$, $VCC2_13$, $VCC2_14$, $VCC2_15$, $VCC2_16$, $VCC2_17$, $VCC2_18$, $VCC2_19$, and $VCC2_20$.

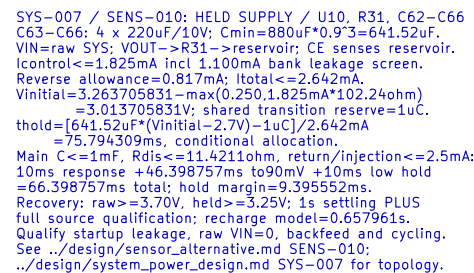
CMS-013; +20mA steady CSI branch planning allocation
4.1mA PHY + 1.989593mA bleed + 13mA allocated GND
+ 0.102mA SET + 0.100mA auxiliary = 19.291593mA.
13mA is an engineering envelope, not an interpolated guarantee.
Auxiliary allowance requires verification; startup/inrush excluded.
AVCC/MPI remains within existing MCU allocation.
Main subtotal: 2.075 + 0.020 = 2.095A; full system budget pending.
At 3.3V: 2.095A @ 3.3V = 6.9135W. Total system efficiency:
Input: 2.961817158A; converter loss = 3.369453727W
Source, current and thermal acceptance remain open.
Sources, conditions and Python: design/camera_storage_interfaces.md

RA8P1 e-reader		
Sheet: /RA8P1 IO and analog supplies/ File: mcu_supply_power.kicad_sch		
Title: RA8P1 I/O and analog supplies		
Size: A3	Date: 2026-09-05	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.6		Id: 4/14





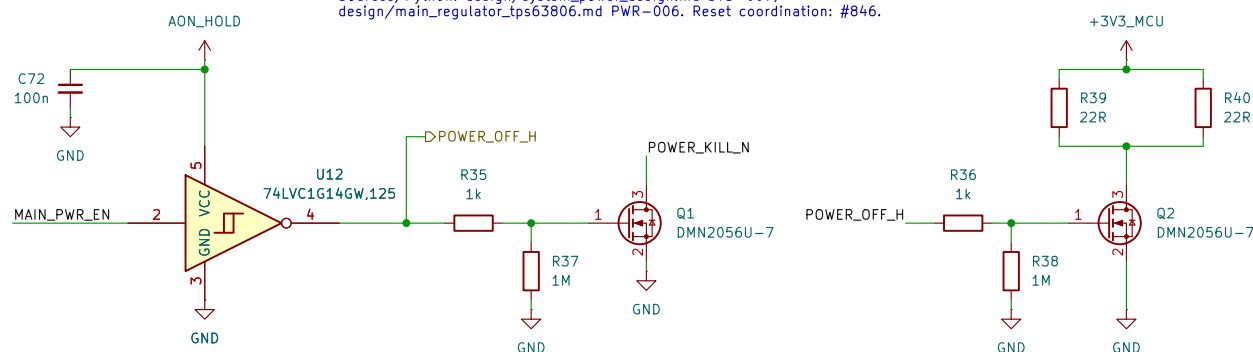
ERC supply path: U10 VOUT through R31.
PWR_FLAG declares that series-fed source only;
raw SYS_AON source remains unimplemented.

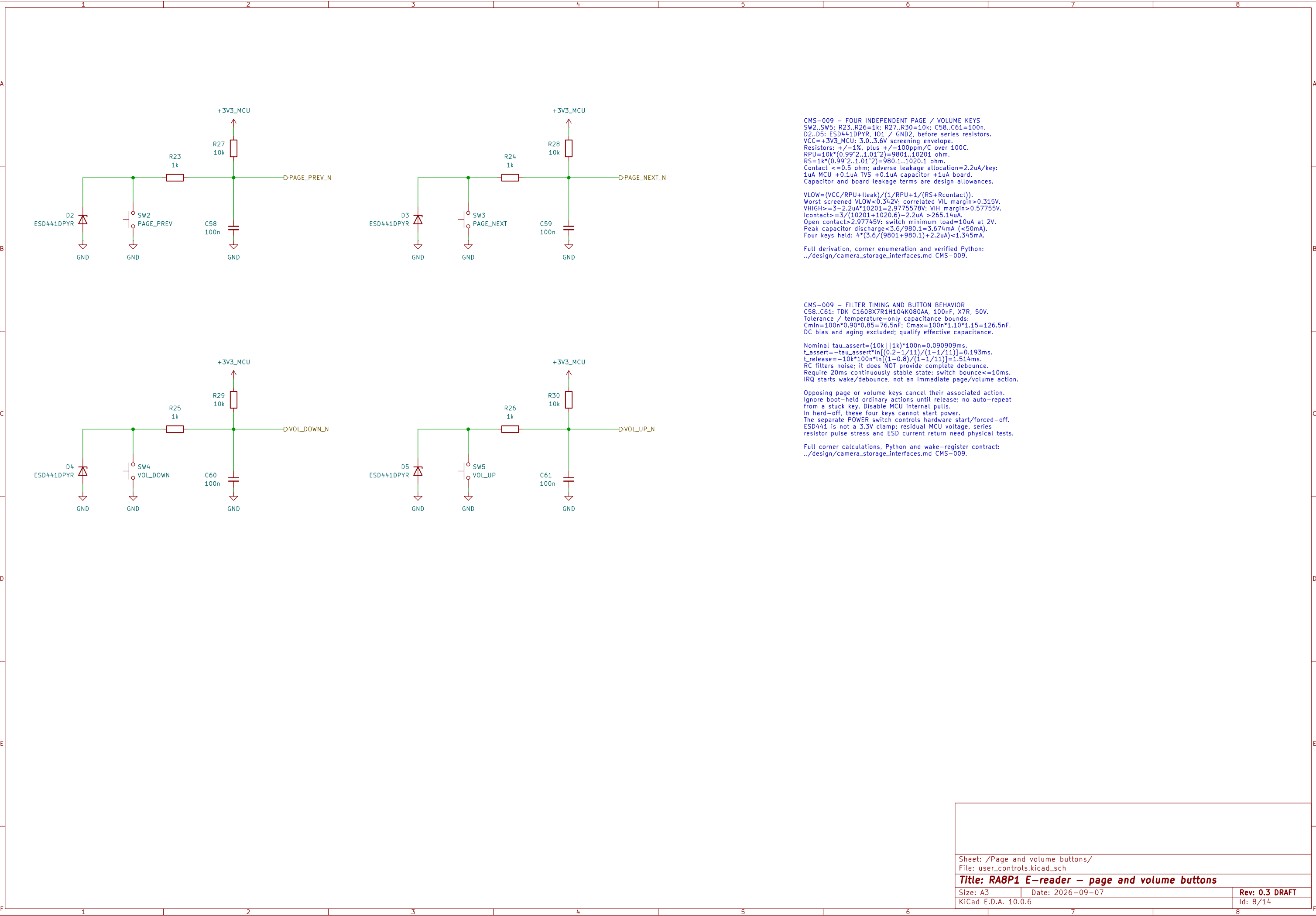


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SYS=007 / PWR=0006: SOURCE SUPERVISION / U11, R32-R34, C67-C71
Vtrips=Vref*(1+R32/R33)+Isense*R32.
Nominal=0.405*(1+732k/100k)=3.369600V.
Vref +/-%2, Isense +/-%25nA; each R +/-%0.1, 25ppm/C, 100C.
Corner trips=3.263705831...3.476597632V.
C68: tau=(732k|100k)*1n=807.980769ns (nominal).
C69-C71: 3x100n COG; initial/temp=284.145, 316.827nF.
CT delay=300ms*1.174+0.0005=1.714786ms nominal.
0.6*(284.145/175+0.0005)=0.974511s MODEL, not guarantee.
Require qualified reset hold >=0.90s >650ms KILL blank +10ms.
R34=110k; ENhigh>=1.7-3.8uA*110k*10^12=2.273559V.
P0.0 sink (1.0-2.0)uA*100k=0.3-0.8uA=14.000000mV.
MAIN_PWR_EN_driver=1.7-3.8uA buffer.
Q3 clamps TP563806 EN.
Held U12 also drives KILL/discharge: USB-clear still needed.
Current EN network and rail calculations: PWR=006.
Sources/Python: design/system_power_design.md SYS=007:
design/main_regulator_tps63806.md PWR=006. Reset coordination: #846.

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CMS-009 – FOUR INDEPENDENT PAGE / VOLUME KEYS
SW2..SW5; R23..R26=1k; R27..R30=10k; C58..C61=100n.
D2..D5: ESD441DPYR, IO1 / GND2, before series resistors.
VCC=+3V3_MCU; 3.0..3.6V screening envelope.
Resistors: +/-1%, plus +/-100ppm/C over 100C.
RPU=10k*(0.99^2..1.01^2)=9801..10201 ohm.
RS=1k*(0.99^2..1.01^2)=9801..10201 ohm.
Contact <=0.5 ohm; adverse leakage allocation=2.2uA/key:
1uA MCU +0.1uA TVS +0.1uA capacitor +1uA board.
Capacitor and board leakage terms are design allowances.

$VLOW=(VCC/RPU+Ileak)/(1/RPU+1/(RS+Rcontact))$.
Worst screened VLOW<0.342V; correlated VIL margin>0.315V.
VHIGH>=3-2.2uA*10201=2.9775578V; VIH margin>0.57755V.
Icontact>=3/(10201+1020.6)-2.2uA >265.14uA.
Open contact>2.97745V; switch minimum load=10uA at 2V.
Peak capacitor discharge<3.6/980.1=3.674mA (<50mA).
Four keys held: 4*(3.6/(9801+980.1)+2.2uA)<1.345mA.

Full derivation, corner enumeration and verified Python:
../design/camera_storage_interfaces.md CMS-009.

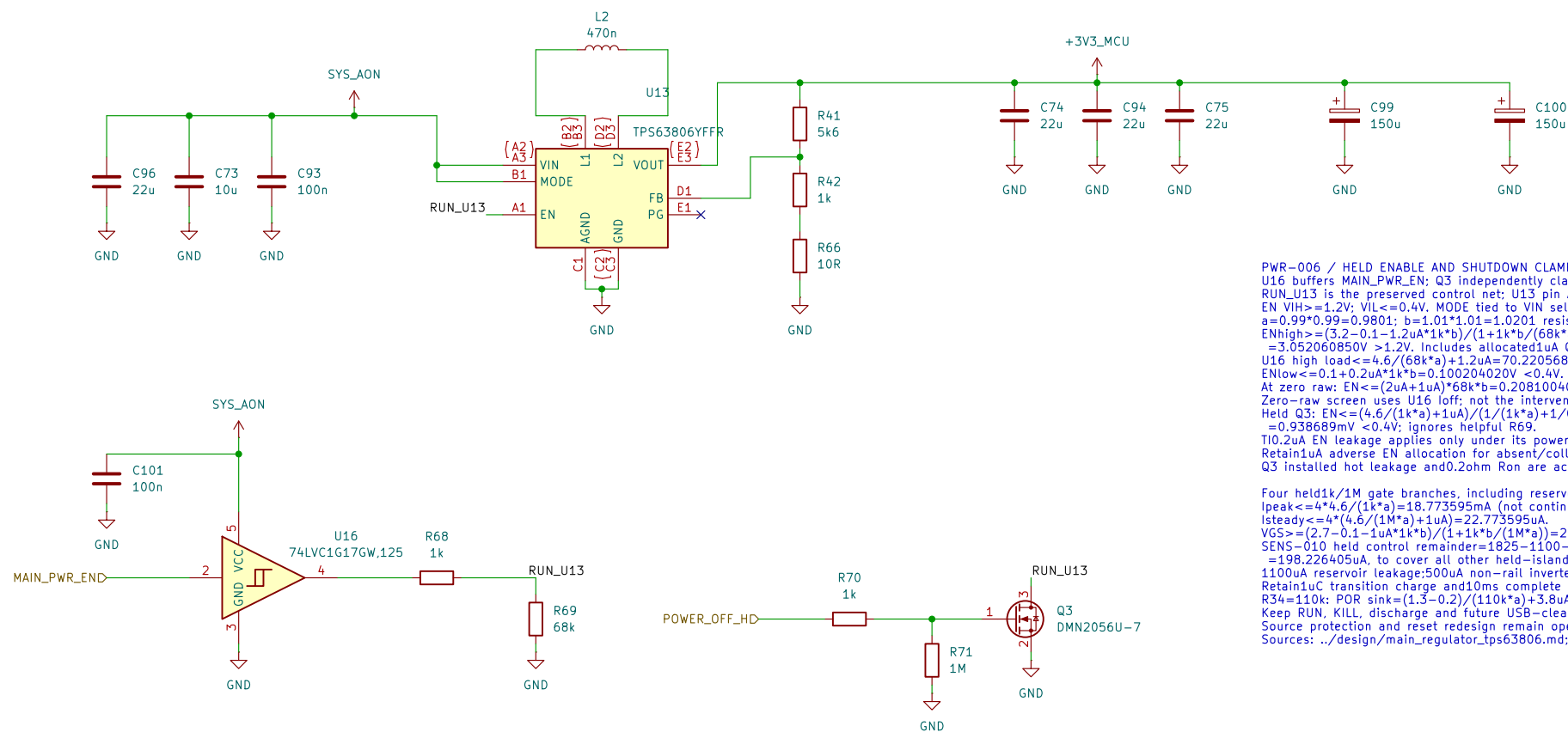
CMS-009 – FILTER TIMING AND BUTTON BEHAVIOR
C58..C61: TDK C1608X7R1H104K080AA, 100nF, X7R, 50V.
Tolerance / temperature-only capacitance bounds:
Cmin=100n*0.90*0.85=76.5nF; Cmax=100n*1.10*1.15=126.5nF.
DC bias and aging excluded; qualify effective capacitance.

Nominal tau_assert=(10k||1k)*100n=0.090909ms.
t_assert=-tau_assert*ln[(0.2-1/11)/(1-1/11)]=0.193ms.
t_release=-10k*100n*ln[(1-0.8)/(1-1/11)]=1.514ms.
RC filters noise; it does NOT provide complete debounce.
Require 20ms continuously stable state; switch bounce<=10ms.
IRQ starts wake/debounce, not an immediate page/volume action.

Opposing page or volume keys cancel their associated action.
Ignore boot-held ordinary actions until release; no auto-repeat
from a stuck key. Disable MCU internal pulls.
In hard-off, these four keys cannot start power.
The separate POWER switch controls hardware start/forced-off.
ESD441 is not a 3.3V clamp; residual MCU voltage, series
resistor pulse stress and ESD current return need physical tests.

Full corner calculations, Python and wake-register contract:
../design/camera_storage_interfaces.md CMS-009.

MAIN DIGITAL SUPPLY: TPS63806 / PWR-006
3.272277 V nominal; forced PWM; held shutdown clamp.
Divider and bypass calculations: design/main_regulator_tps63806.md
Reset coordination (#846), source protection and qualification remain open.
DEVELOPMENT SCHEMATIC – NOT FOR POWER-UP OR FABRICATION.



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PWR-006 / HLD ENABLE AND SHUTDOWN CLAMP
U16 buffers MAIN_PWD_EN; Q3 independently clamps EN during raw collapse.
RUN_U13 is the preserved control net; U13 pin A1 is now EN.
EN VIH>=1.2V; VIL<=0.4V. MODE tied to VIN selects forced PWM.
=0.99+0.99*0.9801; b=1.01*1.01+1.0201 (resistor factors)
= (3.2-0.1-1.2uA)*1k*b/(1-0.99)
= 3.052060850uA. Included allclocks_U13 off leakage.
U16 high load = 4.6/(68k*1u) + 1.2uA = 70.220568uA <100uA.
ENlow = 0.1+0.2uA*1k*b = 0.100204020V <0.4V.
At zero raw: EN<=(2uA+1uA)*68k*b = 0.208100400V.
Zero-raw screen uses U16 loff; not the intervening undefined region.
Held Q3: EN<= (4.6/(1k*1u)+1uA)/(1/(1k*1u)+1/0.2ohm)
= 0.9386mV <0.4V; ignores held Q3.
T10.2uA EN leakage applies only under its powered test conditions.
RetainU1A adverse EN allocation for absent/collapsing raw.
Q3 installed hot leakage and0.2ohm raw are acceptance allocations.

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Four held1k/1M gate branches, including reserved USB-clear:
Ipeak<=4*(4+6/(1k^a))=18.773595mA (not continuous ratings).
Isteady<=4*(4+6/(1M^a)+1A)=22.773595mA.
VG5=(2.7-0.1/(1A+1k^b))*(1+1k^b/(1M^a))=2.566278V.
SENS=high-level control, remainder=862.5mA=22.773595
=198.224650uA, to cover all other held-island loads.
1100uA reservoir leakage;500uA non-rail inverter-input allocation.
RetainLTC transition charge and10ms complete response allocation.
R34=110k; POR sink=(1.3-0.2)/(110k^a)+3.8uA=14.003041uA.
KILL=KILL pin, no charge and no current source separate.
Source protection and reset redesign remain open; no power-up approval.
Sources.../design/main_regulator_tps63806.m; SENS=010 revision.

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PWR-Vom / FEEDBACK AND LOCAL PASSIVES
 $V_{nom} = 0.500 * (1 + 5600 / (1000 + 10)) = 3.72727228 \text{ V}$
R41: 0.02%, 5ppm/C; R42: 0.01%, 2ppm/C; R66: 1%, 200ppm/C.
Installed boards: initial "TCR(100C)"aging assembly, then ohms.
 $R41 = 5600 * (1 + (-0.0002)) * (1 + (-0.0005)) * (1 + (-0.001))$
 $* (1 + (-0.0005)) +/- -0.02 = 5587.6629237, 5612.349243 \text{ ohm}$
 $R42 = 1000 * (1 + (-0.0001)) * (1 + (-0.0002)) * (1 + (-0.001))$
 $* (1 + (-0.0005)) +/- -0.01 = 998.190970, 1001.810970 \text{ ohm}$
 $R66 = 10 * (1 + (-0.01)) * (1 + (-0.02)) * (1 + (-0.01)) +/- -0.05$
 $= 9.954980, 10.455020 \text{ ohm}$ once combined aging/assembly 1%.
Vr=1.000*(1+41.9680/(R41min-R41max))+1.000*(1+1.000/R41max)
Vrl=0.505*(1+41.9680/(R42min-R66min))+1.00nA*R41max
Vstatic=3.2268196m,3.318012496 V (FPWM only).
Once combined +/-75mV regulation/ripple/transient/routing allowance:
Vcomplete=3.315189680..3.393012496 V: qualification required.
SDRAM static VOH margin=2.4-0.743.393012496v=24.891253mV.
This is not measured S/I/noise overhead or reset qualification.

12: 470nH +/-20%; catalog spread alone is insufficient.
Screen incoming0.412..0.518uH; allocated operating+/-10%
= +/-3708..0.5698uH versus T1037..0.57uH. Quality hot/current.
C74/C75/C94: *322uF*0.8*0.85*0.6=6.2928uF > T121uF
C96: 22uF*0.8*0.85*0.6=8.976uF > T14uF input minimum.
The capacitor combined resistance is not guaranteed.
Samsung typical output bias retention at3.393012496V-90.221554%.
Input bias, aging, mounted impedance and startup need verification.
Full assumptions, sources and executable Python: click this note.

PWR-006 / LDA, STORAGE AND SHUTDOWN
Allocation: 1.8A operating; 2.25A cold start screen, not measured load.
Pwr=3.393012496e+1,8=6.107424932E+3W
Thermal allocated Tjmax efficiency: Pin=84.33229991W; lin@3.2V=2,544759372A
Losses = 2.6807876W; loss = 0.678602499W
Tj90% efficiency: loss=0.678602499W,
Tj screen=70C+78.8C/0.678602499W=123.474C.
2.5A board theta-JA is not final PCB thermal resistance.
Pin=3.393012496e+1,8=6.107424932E+3W
Converter limit, source limit, thermal and thermal envelope need validation.

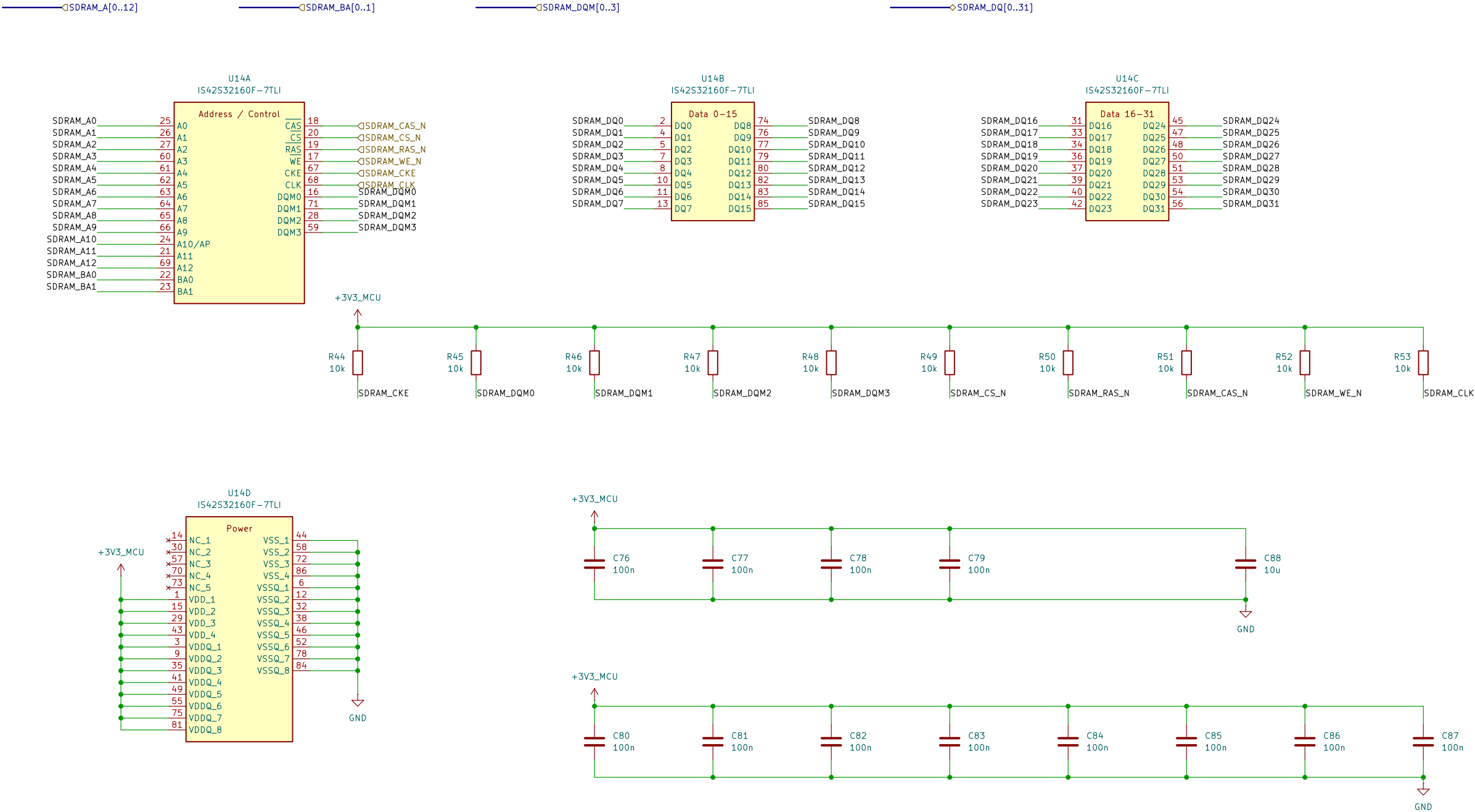
PWR=006 storage checkpoint (nominal, not effective minimum):
 Direct main469.21uF + C43 through FB1 10uF + key filters0.4uF
 =479.61uF cold; radio-on add±0.2uF =>489.81uF.
 C99/C100 remain±150uF each. C94 adds22uF to the output.
 Old bootstrap/private VCC/VyF capacitors were not main-rail storage.
 Whole-main ceiling remains1mF; reconcile later peripheral additions.
 Cout lower acceptance and distributed PDN/startup are separate checks.

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SENS-010 discharge revision: Rmax1x1.42110hm: return/injection<=+2.5mA.
Includes1.5mA key return +1mA other backfeed; Vinf=0.028552750V.
t=6F.91x1.42110hm*ln((3.6-Vinf)/(0.09-Vinf))+10ms+10ms
=66.3987575ms total, below475.794309ms screen.
Includes10ms delay+10ms delay/lamp/discharge+10ms delay/lamp.
Held budget:1.825mA control +0.817mA reverse;IUC transition reserve.
198.226405uA control remainder must cover other held-island loads.
SYS_AON source protection, loaded startup and fast faults remain open.
RST=002 / RADIO=019: release margin remains conditional on rail/path bounds.
Full power per rail path, full current, and control and injection are complete.
Full assumptions: ./design/main_regulator_tps63806.md
shutdown revision/Python: ./design/sensor_allocator.md SENS-010.

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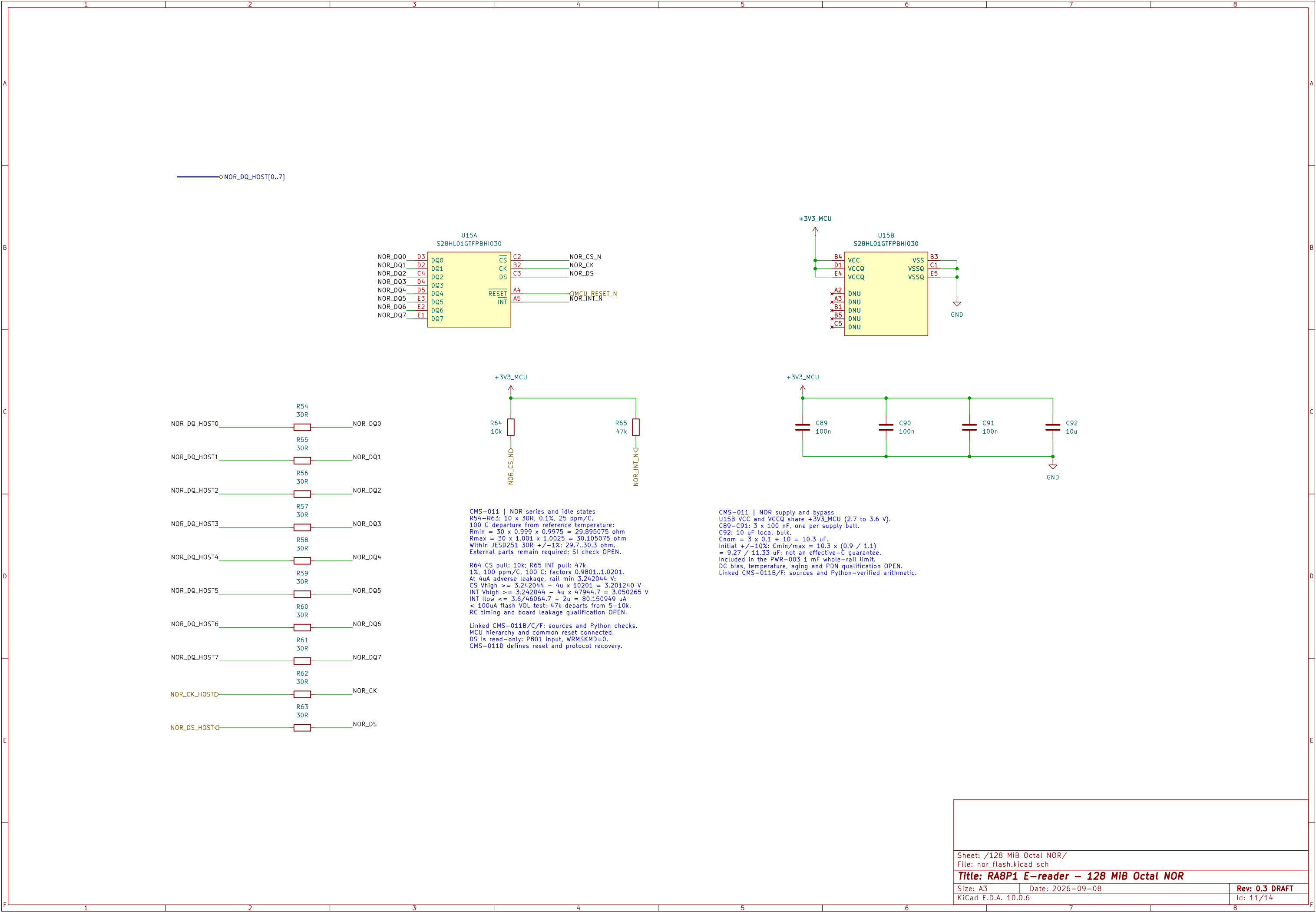
64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset—default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.

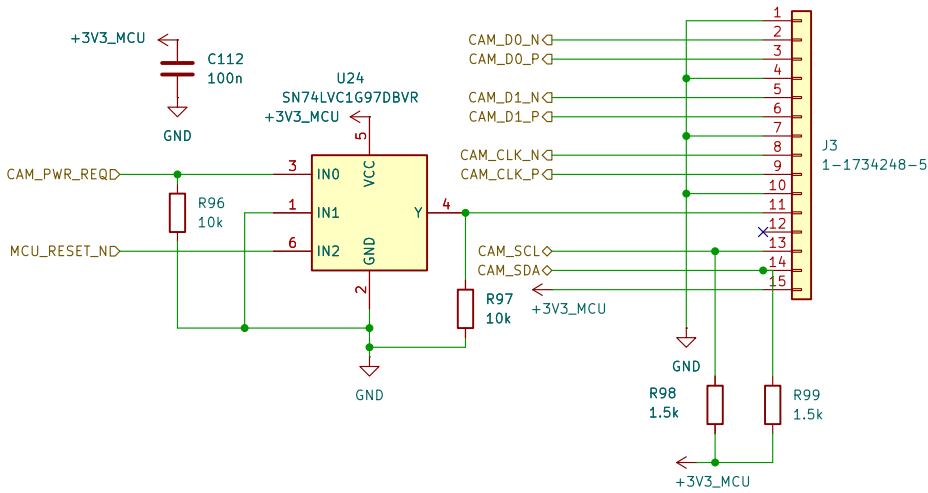


CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44-R53: 10k reset defaults
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT|<=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

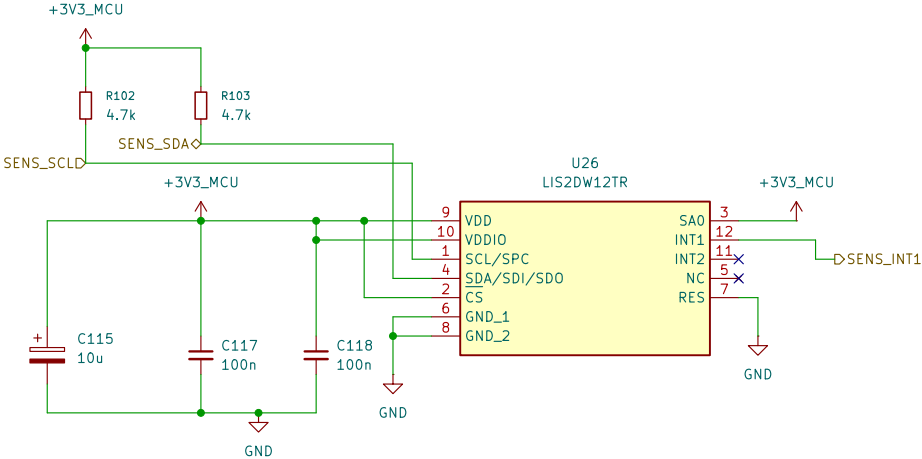
CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail;
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.





CMS-014 / CMS-015: Pcam 5C, 2-lane CSI-2 to VIN0
U24: PWUP = CAM_PWR_REQ AND MCU_RESET_N.
PWUP low >=100ms; then high >=50ms before SCCB.
Qualification and second-camera/lighting work remain open.
See ../design/camera_storage_interfaces.md, CMS-014..016.

ORIENTATION SENSOR – WIP
U26: LIS2DW12TR; ST DS11811 Rev 9 pin map.
VDD pin 9 / VDDIO pin 10: +3V3_MCU; CS / SA0 high for I2C address 0x19.
C117/C118: local 100n bypass; GND pins 6/8 and RES pin 7 grounded.
C115: 10u/16V aluminum EEE–FM1C100R; place near VDD pin 9 with C117.
R102/R103: 4.7k SCL/SDA pull–ups; obsolete ADXL VREG network removed.
Host connected: SCL P410/K13; SDA P409/R16; INT1 P306/D11.
Wait 20ms after supply valid. Power–cycle below 100mV for >=10ms.
Pending: sourcing fields, pin modes, timing/power/passive qualification.
See design/sensor_alternative.md; circuit qualification remains open.



Sheet: /Orientation sensor/
File: orientation_sensor.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.6

Rev:

Id: 14/14